USER REQUIREMENTS SPECIFICATION

Integrated Projecting Environment (IPE)

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Reviewers

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1. Introduction

Improved business systems, productivity and quality of outcome are essential to maintain our market position. The CSIA initiative places significant focus on improving our business systems and quality of outcome. However, productivity improvements will be limited.

Improving the tools available to our engineers and project managers to do their work presents significant opportunity to improve our productivity and competitiveness in the market. Standard tools and methods will also improve our quality of outcome.

1.1. Background

"Tools" is a generic term for the processes, utilities and systems that we have developed to help our staff do there work efficiently.

CTEK has a number of tools that we use to engineer automation projects. The use of these tools is inconsistent and the ability of our staff to fully utilise them varies significantly. However, the use of tools is inherent in how we engineer all the time. The tools tend to be standalone in nature with minimal integration.

Currently these tools focus on "code" generation from Grafcet format Detailed Design Specifications (DDS's), with a strong influence from Fonterra's specific documentation requirements.

Also, our tools are focused on the early development stages of a project. At late stages they don't help much and may even hinder productivity.

The journey to date (CSE to CTEK) has occurred over a period of 17 years with around 16 of those years involving some level of development work on our tools. Typically this development has been reactive to specific project needs and ad hoc in nature.

1.2. Current Situation

The Technical group over the last couple of years has investigated some 3rd party "S88" orientated development environments, with the hope that one of these may suit our requirements. We found that these tools do not fit well with our requirements, although they each offered some great features.

The number of products publically available was two. One suspects that if companies have tools to improve their efficiency, they haven't built them to sell to competitors and therefore you don't get to find them.

Additionally, they typically come from companies of similar or smaller size than CTEK. Initial discussions with these companies indicated that we would have limited ability to influence the functionality of these products. Either because of the reluctance of the supplier and/or the limited resource they could apply to the changes. This was of significant concern, as to adopt one of these products would mean that a significant portion of all of our automation work would rely solely on these tools having the appropriate functionality.

1.3. Path Forward

We have therefore focused on reviewing the functionality of our current tools. At this initial stage it is felt that an Integrated Projecting Environment (IPE) will provide the best outcome. The IPE will represent a quantum shift for CTEK: in what we expect from our tools, how we go about engineering a project and will require significant user training for <u>all</u> our engineers, no matter there level of experience or expertise.

Preliminary analysis has indicated that there is the potential to reduce engineering costs in the range of 15% to 25% by putting an Integrated Projecting Environment in place. The reduction will come from lots of small improvement across all the facets/activities/stages of a projects execution. We are looking to make improvement across 100% of the work effort. Our current tools only focus on 14% of the total work effort.

The development of this User Requirements Specification (URS) is a very important first step to ensure that we as a company truly understand what it is that we want from our tools. And that the investment will deliver the predicted cost savings.

This document and others to follow will also assist us to engage fully with the "Users". The "Users" are all the engineers/project managers that will use the IPE to execute their work, and the CTEK management team. Additionally, it is paramount that the IPE User requirements reflect the requirements of the Users. Their engagement and acceptance of a Integrated Projecting Environment is fundamental to its success.

1.4. Glossary

APC	Advanced Process Control
CIP	Clean In Place
DDS	Detailed Design Specification
FRS	Functional Requirements Specification
FDS	Functional Design Specification
GUI	Graphical User Interface
HMI	Human Machine Interface
HRS	Hardware Requirements Specification
IPE	Integrated Projecting Environment
MCC	Motor Control Centre
OEM	Original Equipment Manufacturer
P&ID	Process and Instrumentation Diagram
PCS	Process Control System
PLC	Programmable Logic Controller
PDS	Procedural Design Specification
SRS	Software Requirements Specification
URS	User Requirements Specification

1.5. Document Highlighting

Items highlighted in yellow are items that have not been completed, and are required to be provided as part of the detailed design stage.

Items highlighted in blue are references that require further confirmation.

Items highlighted in green are references that require further review by author.

2. Business Requirements

Item	Requirement		Priority (Core, 1, 2)
U_BUS_01	We require our specification, design, development and testing environment to be independent of the specific requirements of all clients, industries and suppliers.		Core
U_BUS_02	We require our client and supplier specific requirements to be specific implementation requirements only.		
U_BUS_03	The IPE is to ensure that our tools and projecting methodologies are aligned through all phases of a projects lifecycle, from initiation to close-out.		
U_BUS_04	The IPE is to ensure that quality and quality controls are inherent in the IPE such that risk of rework is minimised and a high quality outcome is realised for us and our clients on every engagement.		
U_BUS_05	The IPE is to ensure that the most efficient methods currently available are used by all staff on all projects all the time.		
U_BUS_06	The IPE is to promote the "CTEK way" of project delivery and outcome, no matter who the delivery team is, on every engagement.		
U_BUS_07	The IPE is to ensure gearing off previous works through all stages of a project is maximised, on all projects, where this is the most effective methodology.		
U_BUS_08	The IPE is to ensure that the CTEK development V-Model methodology is utilised. Top down design and bottom up testing.		
U_BUS_09	The IPE is to ensure that our projecting methodologies are integral to the IPE. Project Management, Projecting Lifecycle and Quality Assurance.		
U_BUS_10	The IPE must support many different clients, industries, implementation platforms and the required deliverables (inputs/outputs).		
U_BUS_11	The IPE must support the certainty of change in delivery requirements over time.		
U_BUS_12	The IPE must inherently support the training of current and new users. This capability must be built in to the IPE from the start.		
U_BUS_13	The IPE must present as a single integrated projecting environment.		
U_BUS_14	The IPE must provide the most efficient and user friendly development environment for our engineers.		
U_BUS_15	The IPE shall be adaptable to future improvements in engineering methods		
U_BUS_16	The IPE must provide a single repository for all components, data and information to support & facilitate all aspects of a project from initiation to close out.		
U_BUS_17	The IPE must provide an Integrated Quality Assurance system with automated generation of all peer review, testing and verification documentation.		
U_BUS_18	The IPE must be Scalable from a small to very large project.		
U_BUS_19	The IPE must be fully supported by integrated help.		
U_BUS_20	The IPE must be fully supported by training materials and training.		
U_BUS_21	The IPE must be easy to use via a friendly and intuitive GUI environment.		
U_BUS_22	The IPE must be portable and distributable. From head office, to remote offices, to site and to users residences.		
U_BUS_23	The IPE shall facilitate collaboration among team members, whether they are located locally or not.		
U_BUS_24	The IPE shall gear off 3rd party products to achieve requirements so as to achieve the best return on investment and functionality trade-off. Consider for each and every element of the IPE.		
U_BUS_25	The IPE shall support access to existing "tools".		

3. Structural Overview

3.1. IPE and Projecting Methodology Overview

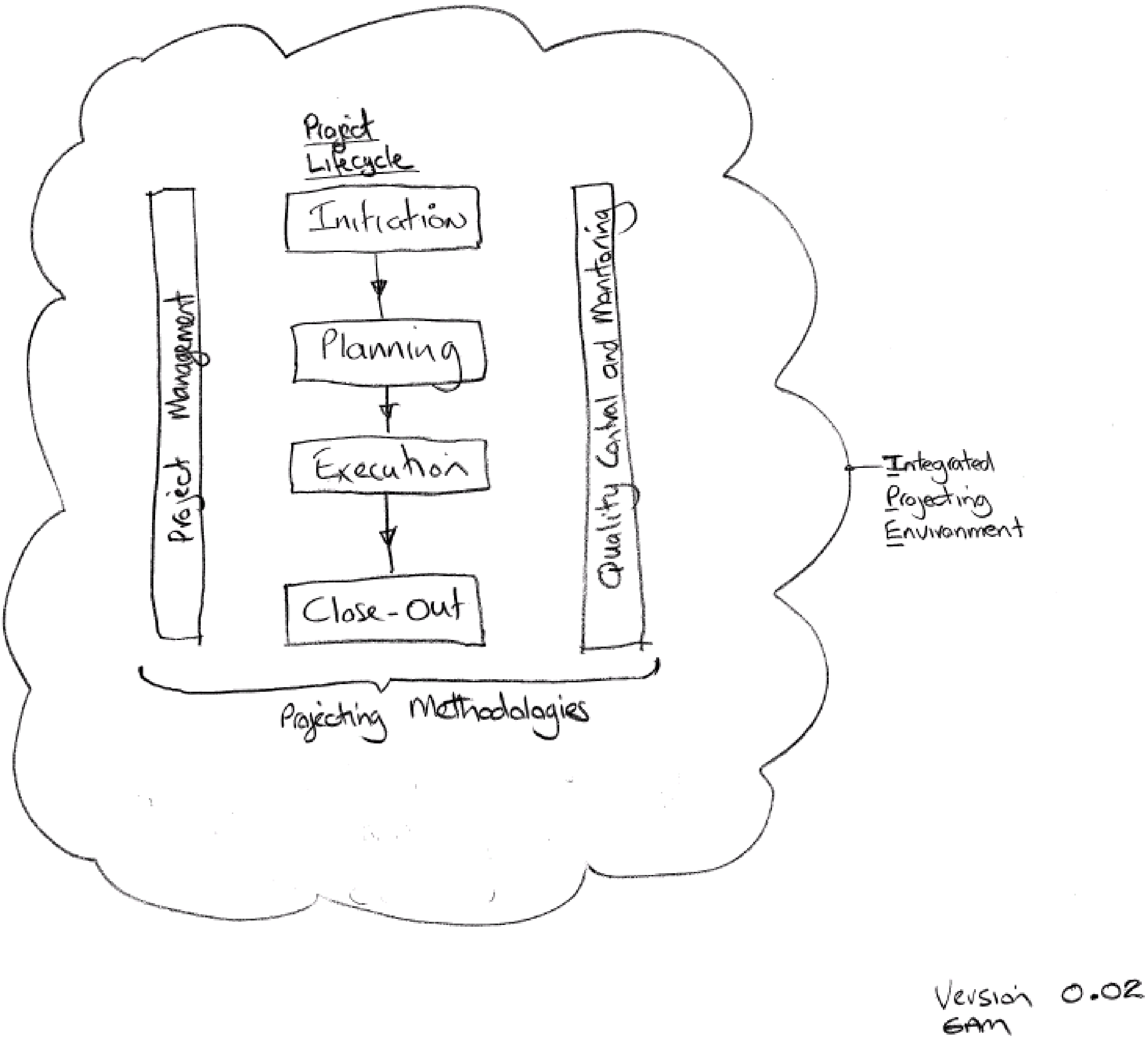

Figure: Integrated Projecting Environment and Projecting Methodology Overview

3.2. Integrated Projecting Environment Overview

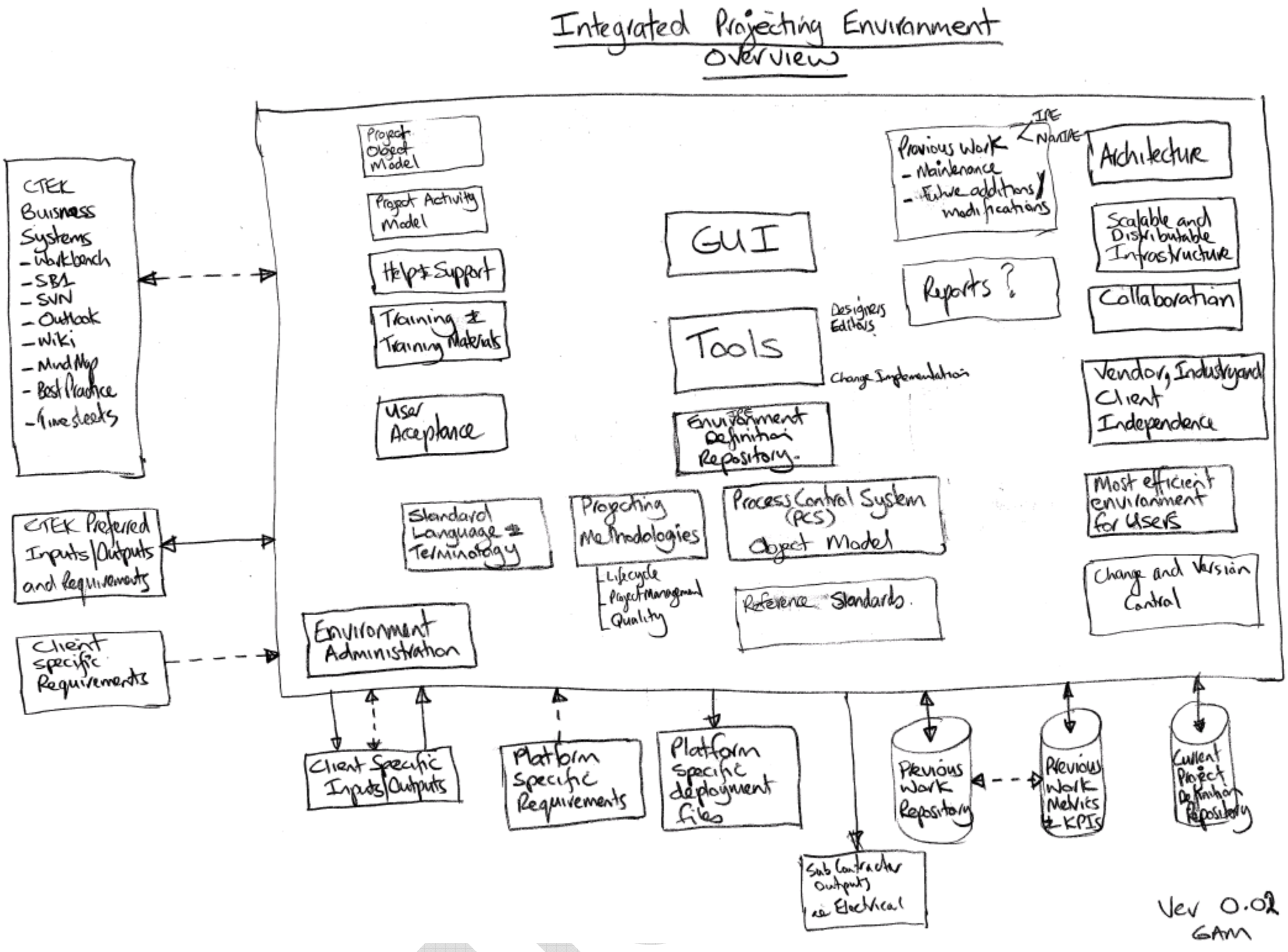

Figure: Integrated Projecting Environment Overview

4. Cornerstone Requirements

The requirements that follow are components/items that must be in place to enable the IPE to be developed. They contain the foundation principal's upon which the IPE will be based. Additional specifications may be necessary to facilitate their development.

4.1. Excel Grafcet environment

The current Excel Grafcet Detailed Design Specification environment shall remain in place to facilitate the use of the large number currently in use by our clients. A review of this environment is required to determine the path forward. The use of this environment will be limited to situations where the client can not be convinced to migrate to IPE.

4.2. Standard Language and Terminology

A standard language and set of terms (terminology) is required to help describe a projects lifecycle, methodologies and object models. It is also fundamental that the language and terms used to describe a Process Control System (PCS), the plant to be controlled and the implementation are standardised.

4.3. Standards

The IPE and our projecting methodology shall draw on the principles contained in various international standards and specific company standards. These standards shall be formally identified and documented.

4.4. Projecting Methodology

A set of standard methods are required for conducting projects. Much of this already exists, as institutional knowledge within CTEK. These methods shall be formally documented and be available within the IPE. The IPE's structure and function shall be based on these methods. The key methodology documents will be Project Lifecycle, Quality Assurance, Project Management, and Engineering Methods.

4.5. Object Model(s)

Standard Object Model(s) shall be created to represent a Project and a Process Control Systems Definition. The IPE's structure and function shall be based on these methods.

In this case the system we are modelling is the entire project lifecycle of a PCS, from first contact (the initial inquiry regarding a potential project) through requirements development (pre tender phase), tender/quotation/estimating, project initiation, project execution, handover to post project support. At all stages the system model includes interaction with other CTEK business systems.

4.6. Project Activity Model

All the activities (tasks/actions) that could be undertaken during a projects lifecycle shall be documented. This will ensure that we understand and cater for all project activities within the IPE. This will assist the building of the "use cases" for the functional specification steps of the IPE development.

4.7. Plant Modelling

A standard methodology for modelling plants shall be developed and documented. This methodology shall draw on international guidelines and practices.

5. IPE User Requirements

5.1. Guiding Principles

Requirements applicable to all aspects of the IPE

Item	Guiding Principles - Requirement	Source	Priority (Core, 1, 2)
U_GP_001	The primary purpose of having an IPE is to make executing PCS projects Faster, Better, Cheaper. Therefore the IPE shall be designed to be more productive to use than not, at all stages of a PCS project.	U_US1_040	
U_GP_002	The IPE shall be designed to support and facilitate executing a PCS project according to CTEK standard Project Lifecycle methods and practices.	U_US1_001 U_US2_002 U_US2_015 U_TCH_032	
U_GP_003	The IPE shall be designed to support and facilitate CTEK standard methodologies, including • Projecting methods • Quality system • Modelling methodology • Standard engineering methods • Project workflow • Adopted international standards	U_US1_014 U_US1_025 U_US2_001 U_US2_033 U_TCH_031 U_TCH_032 U_TCH_034	
U_GP_004	All data contained in the IPE shall conform to a set of overriding criteria:		
U_GP_005	• Any given piece of data shall be entered only once. It is then accessible in all relevant contexts. Change to a piece of data is to reflect into all locations where that data is referenced	U_US1_009 U_US1_023 U_US2_025 U_US2_026 U_US2_042	
U_GP_006	• Data store is common to all users, so the same data is available to all users	U_US1_002 U_US1_008 U_US2_027	
U_GP_007	• Once entered all data is to persist for the life of the project. Any editing of data is stored as changes to that data, so that change history is available for all edits.	U_US1_010 U_US2_087 U_US2_037	
U_GP_008	• Data version control and backup shall be integrated into the IPE	U_TCH_026 U_TCH_028	
U_GP_021	• IPE to be aware of PLC code, so that any manual change to code is detectable, and may be accepted or rejected for inclusion into the IPE model	U_US3	
U_GP_009	The IPE shall be Easy To Use. This requirement is essential, and is to be reflected in all design decisions in building the IPE	U_US1_037 U_US2_068 U_TCH_016	Core
U_GP_010	The IPE shall incorporate user based security, in order to: • Protect CTEK's investment from loss/theft • Provide a customised user experience depending on the users role • Log activity against users • Register a users interest in a particular aspect of the PCS model	U_US1_049 U_US1_052 U_US2_075 U_US2_078 U_TCH_019	
U_GP_011	The IPE shall support project teams of various forms: • Single user, small project • Small teams on small • Large teams on large projects • Collocated teams • Geographically separated teams	U_US1_050 U_US1_052 U_US1_093 U_TCH_018	

Item	Guiding Principles - Requirement	Source	Priority (Core, 1, 2)
U_GP_012	The IPE shall be able to be used in various locations, including • Office • Remote office • Home • Client site, with continuous internet connection • Client site, without continuous internet connection	U_US1_050 U_TCH_018	
U_GP_013	The IPE shall promote cooperation and communication among team members. This includes notification of one user when a module of interest is modified by another user.	U_US1_054 U_TCH_019	
U_GP_014	The IPE shall support progressive migration of a project from office based development to site based installation and commissioning.	U_US1_050	
U_GP_015	The IPE shall be effective of projects of all sizes. That is, it must scale from the smallest to largest projects and provide performance and features appropriate to those projects. This includes keeping the setup effort low so as to minimise the barrier to use on small projects	U_US1_093 U_US2_020 U_US1_094	
U_GP_016	The IPE shall include a Lexicon defining a common language to describe a PCS and its components	U_TCH_030	
U_GP_017	IPE shall include templates for CTEK preferred input data formats (eg IO schedules) for use by customers who do not have a preference in this regard.	U_TCH_024	
U_GP_018	IPE shall include a DDS template (the existing DDS Excel template) for use by customers who do not have a preference in this regard.	U_TCH_029	
U_GP_022	And by customers with an existing set of DDS's	U_US3	
U_GP_019	To facilitate initial and ongoing development of the IPE, diagnostic functions shall be built into the IPE: • Audit trail of IPE execution (record of actions taken by the IPE) • Exception trapping (perhaps use windows event log) • Verbose mode to provide great detail on a repeatable problem	U_TCH_012	

5.2. Project Management

Requirements to support the management function

Item	Project Management - Requirement	Source	Priority (Core, 1, 2)
U_PM_001	IPE is to support use of CTEK project lifecycle management practices.	U_US1_012 U_US1_099 U_US1_066 U_US2_023 U_US2_024	
U_PM_002	CTEK project methodology mandates P&IDs as a critical document. P&ID document status is to be tracked and release for use in PCS modelling is required based on approved status. P&IDs are a controlled document; all modelling based on P&IDs is tagged to drawing revision. Document status's • As Tendered • Under Review • Released • As Built	U_US2_048 U_US2_051	
U_PM_003	IPE to include concept of a project team and roles within the team. Team members include not only CTEK staff and contractors, but client stakeholders too.		
U_PM_004	The IPE shall provide a stakeholder register, including role on the project, contact details, and integration to Outlook and timesheets.	U_US2_082	
U_PM_005	IPE to schedule routine team meeting (integrate with Outlook)	U_US2_003 U_US2_079 U_TCH_026	
U_PM_006	IPE to assign and maintain a status of each component in the PCS model. Status to be assigned at levels of • Declared (added to area model) • Defined (named, described, function defined in broad terms) • Specifying (function specification in progress) • Specified (function specification complete) • Detailing (detail specification in progress) • Detailed (detail specification complete, ready for module test) • Coding (code development in progress) • Coded (code complete) • Testing (module tests in progress) • Tested (module tests passed) • Integrating (integration tests in progress) • Integrated (integration tests passed) • Installing (installed in PCS on site, IO checkout in progress) • Installed (installed in PCS on site, IO checkout passed) • Start-up (initial start-up, dry test, wet test in progress) • Ready (ready for production) • InUse (in use for production)	U_US1_061 U_US1_086 U_US2_046	
U_PM_007	Where possible IPE is to adjust status automatically, or prompt owner if uncertainty exists	U_US1_086	
U_PM_008	Status of modules to be further subdivided to components of the modules Eg Phases and EMs are made up of steps and transitions (among other things) these may be initially defined in broad terms in descriptive language, and later refined as detail is added. Different steps and transitions will be at differing stages of detail so will have different status. Individual component status may be inferred from its content. Overall module status can be inferred from the sum of its components.	U_US2_085	

Item	Project Management - Requirement	Source	Priority (Core, 1, 2)
U_PM_009	Status may be rolled back, eg if start-up reveals deficiencies in functional design		
U_PM_010	Project task list Generate from PCS data model a work list representation modules defined in the area model Assign team members to tasks (primary responsibility) Integrate task assignment into team members Outlook task list	U_US1_086 U_US1_090 U_US2_080	
U_PM_011	Project Gantt Generate from the task list, project scope, resource allocation plan and template a timeline in Gantt format. Track status against this Gantt. Maintain history of changes to plan Allow ‘what if’ scenario analysis Analyse module interfaces to determine dependences and bottlenecks, and adjust Gantt to suit	U_US2_088 U_US2_095	
U_PM_050	Estimate size of modules initially from tender draft area model. Refine size estimate as development progresses Red flag modules if size deviates markedly from budget	U_US3	
U_PM_012	Requirements traceability matrix Support mapping of URS/Contract scope requirements to PCS model components.	U_US2_055	
U_PM_013	Risk register Support creation of a risk register, and risk mitigation plan in accordance with CTEK project management methodology	U_US2_081	
U_PM_014	PCS model dependencies As part of the Area Modelling and Function Design steps, the IPE is to analyse the model component dependencies to assist in identifying common areas and bottlenecks to assist in scheduling the project work	U_US2_088 U_US2_095	
U_PM_015	Fault log Support entry tracking and reporting of Action Items (replacement for paper Fault Log and Commissioning Log)	U_US1_084	
U_PM_016	**Project Kickoff**		
U_PM_017	Once a project is awarded, the IPE shall facilitate handover from the Tender team to the Project Team	U_US1_004	
U_PM_018	An early step is elaboration of the tender package into a delivery package for client review and signoff. Minimum delivery package content to be defined as part of CTEK standard methodology. Additional content may be required on some projects based on client requirements.	U_US2_009	
U_PM_019	**Reporting**	U_US2_012 U_US2_045 U_US2_086 U_TCH_006	
U_PM_020	The IPE shall measure key project metrics and kpi’s and generate reports required by the project manager: • Reports may be generated by user or by PCS component or both • Report by Module status eg all modules with incomplete logic • [Specify reports] •		
U_PM_021	The IPE shall measure key project metrics and generate reports required for presentation to CTEK management: • [Specify reports]		

Item	Project Management - Requirement	Source	Priority (Core, 1, 2)
U_PM_022	The IPE shall measure key project metrics and generate reports required by the client. These reports content may vary by client and project: • [Specify reports]		
U_PM_023	The IPE shall produce print outs/reports for the PCS design for internal CTEK use, in standardised format independent of client documentation requirements: • Area Model • Functional design • Detailed design • Device and IO schedules	U_TCH_006	
U_PM_024	**Task management, PM perspective**		
U_PM_025	IPE to include allocation of activities and modules to team members (consider integration into Outlook task list)	U_US2_076 U_US2_080	
U_PM_026	Budget is allocated to assigned modules based on cost model		
U_PM_027	Progression of a project between major stages requires authorisation from the Project Manage and/or Lead Engineer	U_US2_084	
U_PM_028	Cost is assigned to modules based on timesheet entries (time sheet entries may be derived fro IPE activity on a given module)		
U_PM_029	IPE to generate a requirements matrix of User Requirements against Functional Specification	U_US2_055	
U_PM_030	IPE shall organise identified tasks into a schedule (Gantt) honouring identified dependencies. PM may adjust schedule as he sees fit.	U_US2_095	
U_PM_031	**Task management, lead perspective** Lead engineer(s) may perform some PM tasks, depending on size of project team		
U_PM_032	**Task management, user perspective**	U_US2_0	
U_PM_033	**Quality Assurance**		
U_PM_034	The IPE is to integrate CTEK's Quality System, to ensure the produced PCS is of the required quality.	U_US1_100	
U_PM_035	The IPE shall ensure a v-model top down design, bottom up testing methodology, with peer review at key points is followed	U_US2_007 U_US2_010 U_US2_019	
U_PM_051	Where Quality System specifies sign-off gates, these are to be enforced in IPE	U_US3	
U_PM_036	**Client Reporting**		
U_PM_037	IPE to support collection of data and report generation for takeover according to contract specification	U_US1_096	
U_PM_038	IPE to support collection of data and report generation for acceptance according to contract specification	U_US1_096	
U_PM_052	IPE to provide on-line reporting of project status for client to access via web	U_US3	
U_PM_053	IPE to support generation of reports specified in contract, eg weekly task/resource Gantt	U_US3	

5.3. IPE Use in Tender Preparation

Requirements for IPE to assist Tender Preparation

Item	IPE Use in Tender Preparation - Requirement	Source	Priority (Core, 1, 2)
U_TND_001	The IPE shall be used for all stages of a Projects Lifecycle, including initial scoping, project qualification and bid preparation.		
U_TND_002	Initial scoping shall include an assessment of the suitability of using the IPE for the project at hand	U_US1_095	
U_TND_011	Option to use only project management and Quality aspects of IPE if project does not suit rest of IPE	U_US3	
U_TND_003	The IPE shall capture all decisions and assumptions made during initial scoping and design in the bid preparation process. Data entered during this phase shall carry forward to project kickoff and form the basis of the initial PCS model. Data captured to include: • P&IDs • URS and scope documents • IO • Panel layouts • Architecture • Process Flow drawings • Area Model information/drawings. • Client standards • Photos and Video • Design Notes • Meeting minutes • Gearing decisions • Basis of costing decisions • Time estimates • Project Resource estimates • Project Timeline estimates	U_US1_003 U_US1_006 U_US1_007 U_US2_029	
U_TND_004	Revision of the initial design derived from the tender process shall be assessed for 'red flag' deviation that may materially impact the cost or outcome of the project.	U_US2_022 U_US2_029	
U_TND_010	Integrate tender costing process into IPE, including: • Preliminary Area Modelling, IO and module counts • Estimating uniqueness and complexity • Cost modelling • Requirements tracing • etc	U_US3	

5.4. IPE Features

Requirements for specific aspects to be incorporated into the IPE

Item	IPE Features - Requirement	Source	Priority (Core, 1, 2)
U_FTR_001	**Design independence from Client and Vendor requirements**	U_US2_004 U_US2_005 U_US2_013 U_US2_031 U_US2_032 U_US2_061 U_US2_062 U_US2_063 U_US2_064 U_US2_065 U_TCH_017 U_TCH_020 U_TCH_021 U_TCH_022 U_TCH_023 U_TCH_033	
U_FTR_002	The IPE shall provide a design environment **independent** of all client and vendor specific requirements. To support this, the IPE shall:		
U_FTR_003	• Represent the PCS in a CTEK defined standard model		
U_FTR_004	• Provide a range of optional Client and Vendor specific **plug-ins** to map the CTEK model to particular PLC/HMI etc packages and to client code standards		
U_FTR_005	• Present and document the PCS in a CTEK defined standard method		
U_FTR_006	• Provide a range of optional Client specific plug-ins to map the CTEK model to particular client documentation standards		
U_FTR_100	• Provide a range of optional Client specific **plug-ins** to map the client supplied specification documents to CTEK model standards, eg P&IDs, Schedules, Requirements documents	U_US1_074	
U_FTR_007	Include a CTEK standard plug-in set for PLC code, HMI and documentation, for use by clients without their own standards	U_US2_094	
U_FTR_008	**User Training**	U_TCH_013 U_US2_021	
U_FTR_009	The IPE shall incorporate features to facilitate training users in the use of the IPE itself and application of CTEK standard methodologies	U_US1_026 U_US2_069 U_US2_070	
U_FTR_010	Passing basic training modules is a prerequisite to using the IPE	U_US2_070	
U_FTR_011	Training materials shall include a sample project, selected and designed to introduce new employees (especially new graduates) to CTEK methods and the IPE	U_US1_026	
U_FTR_012	Training modules shall be prepared targeted at users of varying experience	U_US2_021	
U_FTR_013	**User Help**		
U_FTR_014	The IPE shall include a comprehensive Help system	U_US1_051 U_TCH_025	
U_FTR_015	IPE Help shall be context sensitive	U_TCH_025	
U_FTR_016	IPE Help shall be searchable	U_TCH_025	
U_FTR_017	IPE Help shall be relevant and comprehensive Ie must not be like "To do <*some unfamiliar term*> press the <*unfamiliar term*> button"	U_TCH_025	
U_FTR_018	IPE Help shall be available in comprehensive and quick start form (suitable for the novice and experienced user respectively).	U_TCH_025	
U_FTR_019	An IPE Wiki shall be included (part of IPE, or part of CTEK Wiki?) to form a knowledge base and user forum	U_US1_027	
U_FTR_101	Users can annotate help with personal notes. These notes to be available to all.	U_US3	

Item	IPE Features - Requirement	Source	Priority (Core, 1, 2)
U_FTR_020	**Team Collaboration Features**	U_US2_014	
U_FTR_021	The IPE shall support multiple users accessing and modifying a model concurrently	U_US1_043	
U_FTR_022	Any component of a IPE model may be viewed may many users simultaneously		
U_FTR_023	Any component of an IPE model may be locked for edit by one user. The lock status shall be visible to all users.	U_US1_053	
U_FTR_024	Where a component is edited by one user and is being viewed by another, the viewing users are to be notified (perhaps by an edits present icon on the view) and may refresh their view	U_US1_054	
U_FTR_025	All components of the IPE model are assigned to a user for development. Only assigned components may be edited.		
U_FTR_026	Users may make suggestions for changes on modules they are not assigned to, for review and acceptance by the assigned user.		
U_FTR_027	Where changes by one user on a module impact on other modules registered to other users, those users are to be notified that they need to review impact on their modules.	U_US1_054	
U_FTR_028	While conducting meetings, the IPE should be available for direct entry of minutes, thoughts and comments. IPE to include a meeting record (recording when where why and who), with minutes including contextual links to related parts of the model This will require an IPE terminal in all meeting	U_US2_096	
U_FTR_029	IPE to support use of 'Interactive Whiteboard' for team modelling sessions. • Use finger as electronic pen • Voice commands	U_US2_038 U_US2_097	
U_FTR_030	**Client and Peer Review**	U_US1_082 U_US1_068 U_US2_029 U_US2_060	
U_FTR_031	Publish IPE model for review by client (perhaps via web page access)		
U_FTR_032	Client may not modify model		
U_FTR_033	Client and peers may make notes and comments on the model		
U_FTR_034	Client must respond to review with Approved, Approved with Comments or Not Approved.		
U_FTR_035	IPE shall retain a snapshot of the model as reviewed, and retain reviewer comments	U_US1_083	
U_FTR_036	Selected project progress and status reports to be available on-line to the client.		
U_FTR_037	**Templates**		
U_FTR_038	A template specifies a standard implementation of a common piece of process. It is defined to specify the design of a process in a general way that can be applied to a range of projects.	U_US1_029 U_US2_016	
U_FTR_039	Templates must cover a range of scopes: • Whole process, covering multiple components of an Area Model (perhaps a Unit, with all contained phases, EMs, CMs etc) • Single module (Phase or EM) • Fragment (common use piece of logic)	U_US1_029 U_US1_073	
U_FTR_040	Templates may be defined as a class, or as an instance		
U_FTR_102	Template to include capability to specify configuration option, including varying number of sub components (eg silo manifold template, specify number of silos when applying template)	U_US3	
U_FTR_103	Having applied a template, retain ability to change easily configuration decisions at a later date	U_US3	

Item	IPE Features - Requirement	Source	Priority (Core, 1, 2)
U_FTR_041	**Drawing Handling**		
U_FTR_042	The IPE shall support use of AutoCAD drawings in .dwg and/or .dxf format		
U_FTR_043	An IPE project shall include a registered set of P&ID drawings. The registered drawing set shall include drawing properties, including: • Drawing number • Drawing revision • Short descriptive name, typically process area of sub area		
U_FTR_044	The IPE shall include a drawing viewer, including all features of a typical basic drawing viewer package		
U_FTR_045	**Reverse Engineer and Legacy Support**	U_US1_039 U_TCH_014	
U_FTR_046	It shall be possible to import an old project from existing DDS's and PLC code, mapping them to an IPE model.		
U_FTR_047	Depending on the quality of the source material, mapping of an existing project to an IPE model will require more of less manual intervention		
U_FTR_048	To support design by third parties without access to the IPE, support shall be provided to import detail design from DDS provided by others in both CTEK format and other formats.	U_US1_103	
U_FTR_049	**IPE system Administration**	U_TCH_015	
U_FTR_050	The IPE security model shall provide user roles varying access rights • Administrator • Manager • Project Manager • Lead Engineer • Engineer • Client • View only In general access right restrictions are to be as light as possible Possibly integrate with CTEK domain security	U_US1_049 U_US2_075 U_US2_077 U_TCH_015	
U_FTR_104	Individuals roles can vary across projects	U_US3	
U_FTR_051	User administration function are required • Add user • Change password • Lock user account (don't allow delete in order to preserve change history) • User group assignment • Allow assigning a user to multiple roles • Allow user with multiple roles to select role context to use. • Administration management of role privileges	U_US2_077 U_TCH_015	
U_FTR_052	The IPE shall integrate with selected CTEK business systems	U_TCH_026	
U_FTR_053	The IPE shall respect clients Intellectual Property. It shall prevent inadvertent or unauthorised use of one clients IP on another's project	U_US1_098 U_US2_030 U_US2_036	

Item	IPE Features - Requirement	Source	Priority (Core, 1, 2)
U_FTR_054	The IPE shall integrate with CTEK Timesheet system • Monitor user activity and fill in pending time sheet for user confirmation • Assign work to sub jobs based on components being modified • Allow for user initiated interruptions (system tray app, to select alternative jobs) •	U_US2_079 U_TCH_026	
U_FTR_055	**Working Out of Office**	U_US1_034 U_US1_035	
U_FTR_056	At times it will be necessary to move part of a project out of the office (to site) while the remainder continues to develop at the office.		
U_FTR_057	If continuous internet connectivity is available, it shall be possible to continue working on the project as if it were collocated.		
U_FTR_058	If continuous internet connectivity is not available, it shall be possible to lock parts of a project to prevent inadvertent conflicting changes.		
U_FTR_059	It shall be possible to periodically synchronise the project, via intermittent internet connection, or email.		
U_FTR_060	It shall be possible to manually synchronise parts of the project (co-ordinated via telephone conversation)		
U_FTR_061	**IPE Security and Protection**		
U_FTR_062	The IPE shall be designed to resist use by unauthorised persons, whether by loss or theft of the software itself, an IPE installed computer, or by unauthorised access to an IPE installed computer		
U_FTR_063	The IPE should not be able to be installed and made operational on a PC or VM image without express authorisation by CTEK		
U_FTR_064	The IPE should not be able to be moved from one PC or VM image to another without express authorisation by CTEK		
U_FTR_065	An IPE installation on a VM image should not be able to be run on an unauthorised PC		
U_FTR_066	**Project Metrics**		
U_FTR_067	Keep set of key project metrics available, even after old project has been archived. • Client details • Process type details • Projects size (IO, CM and module counts) • Architecture outline • Project complexity (counts of key model features) • Project cost, budget and actual •	U_TCH_027	
U_FTR_068	**IPE Architecture**		
U_FTR_069	Integrated version control (consider integrating SVN or GIT). The IPE should implement a version control and data protection strategy, ideally without the basic user even being aware of it.	U_US2_037 U_US2_079 U_TCH_026 U_TCH_028	
U_FTR_070	**Project Documentation to Client Specification**	U_US1_032	
U_FTR_071	Production of PCS documentation (FDS, DDS etc) to client specified format is treated as a system output.		

Item	IPE Features - Requirement	Source	Priority (Core, 1, 2)
U_FTR_072	Mapping of the CTEK standard PCS Model to client specified formatted documents is specified in the documentation plug-in. This mapping is sufficient to satisfy client documentation requirements only and may not map the entire CTEK PCS model.	U_US1_065	
U_FTR_073	Documents may be produced at any time by a 'Print to Excel' or 'Print to Word' or 'Print to <file format>' function.		
U_FTR_074	While the project is in progress, these documents are non-master, non-controlled snapshots.		
U_FTR_075	At project takeover (when document control passes to the client) preference is to retain control of masters in the IPE, and to issue client format documents for client use and version control. Any changes by the client are fed back to the IPE. Any changes by CTEK are done in IPE only (and document re-issued)		
U_FTR_076	Only at project acceptance (when document control passes to the client) are final master documents produced.		
U_FTR_105	**PCS Management**	U_US1_097	
U_FTR_106	IPE to manage PLC tag values by upload/download: • Process variables • Configuration • Bingo tracks • Initial values (where applicable)		

5.5. IPE User Interface

Requirements related to the design and features of the IPE GUI

Item	IPE User Interface - Requirement	Source	Priority (Core, 1, 2)
U_UI_001	The IPE features for designing a PCS shall be largely graphical.	U_US1_044 U_US1_048 U_US2_041 U_TCH_001	
U_UI_002	The IPE is to present an integrated design environment to the user	U_US1_033 U_US1_041	
U_UI_003	Where it makes sense from a usability or efficiency viewpoint a command line type interface for some functions is acceptable		
U_UI_004	The IPE is to implement a consistent user interface across all components, including consistent : • Menu structure • Keyboard short cuts • Mouse actions • Drag and Drop • Select from dropdown list of available options	U_US1_046 U_US1_047 U_US2_017 U_US2_039 U_US2_040 U_US2_053 U_US2_071	
U_UI_005	Tabular views to support Filters	U_US1_045	
U_UI_006	IPE to provide Cross Reference facility to show all places a component is referenced	U_US1_055 U_US2_054	
U_UI_007	IPE to provide Search facility to find all places a component is referenced	U_US1_055	
U_UI_008	IPE to provide Search And Replace facility to find and update all places a component is referenced	U_US1_055	
U_UI_009	Option to preview and edit logic in ladder	U_US1_063	
U_UI_010	Provide visual highlight of incomplete items (eg transitions with invalid syntax logic)	U_US1_064	
U_UI_011	Provide visual indication of component version and status information	U_US1_081	
U_UI_012	The IPE shall include Wizards to guide the user through common tasks, eg: • Create Phase/EM • Generate some code • Correlate ownership • Create a project in IPE • Create a PLC project • etc		
U_UI_013	Module designer to include ability to animate based on running system data	U_US1_102 U_US2_091	
U_UI_014	IPE User Interface to present options/menus etc based on logged in user role. Access to certain features may be restricted for some roles Users with more than one role may choose the view to use	U_US2_072 U_US2_073	

5.6. PCS Design

Requirements related to the design of a PCS

Item	PCS Design - Requirement	Source	Priority (Core, 1, 2)
U_DSN_001	The PCS design is to be defined in a CTEK specified standard model. This model is to be independent of vendor and client requirements.	U_US1_016	
U_DSN_002	The IPE shall support various basic PCS functional design philosophies	U_US1_079	
U_DSN_003	• Sequential step based control (aka Grafcet)	U_US2_089	Core
U_DSN_004	• House of Cards step based control		2
U_DSN_005	• Cascading permissive control		2
U_DSN_006	• Distributed function block (IEC 61499)		2
U_DSN_007	The IPE design environment shall be organised into several 'Designers' focused on one aspect of the PCS design. Designers include:	U_US1_036 U_TCH_002	
U_DSN_008	• Area Model designer		
U_DSN_009	• Module function designer		
U_DSN_010	• Visualisation designer		
U_DSN_011	• Hardware Architecture designer		
U_DSN_012	Additional designers are required for IPE support functions:		
U_DSN_013	• Plug-in designer		
U_DSN_014	• Template designer		
U_DSN_015	• DDS reverse engineering designer		
U_DSN_016	• PLC code reverse engineering designer		
U_DSN_017	**Designer Common Features**		
U_DSN_018	All records in the PCS data model shall have a unique identifier, independent of any other property, including tag name. All references to one record in the PCS data model by another shall be made using the unique identifier. In this way any change to a record will be reflected into all references to that record without further work	U_US1_062	
U_DSN_019	All edits to the PCS data model shall be pending until accepted by the person editing. Pending edits are not applied to the data model (and hence do not impact on other areas of the model) until accepted.		
U_DSN_020	Pending edits may be assessed for what impact they will have on other areas of the model if accepted.		
U_DSN_021	All components in the PCS model shall accept entry of design notes. These notes are free form text, and may include linked reference to other IPE model components or external documents	U_US2_011 U_US2_029	
U_DSN_022	Notes may be marked as completed (or similar) to avoid unnecessary clutter when viewing the model		
U_DSN_023	The IPE shall include a library of templates of common use artefacts	U_US1_028	
U_DSN_024	The IPE shall provide guidance to the user on correct use and methodology	U_US2_018	
U_DSN_025	**Area Model Designer**	U_TCH_002 U_US1_024	

Item	PCS Design - Requirement	Source	Priority (Core, 1, 2)
U_DSN_026	Area model design is often centred on P&ID's. This designer utilises the drawing viewer and adds features to support Area Model design and presentation		
U_DSN_027	The AM designer creates an overlay, associated with but separate a drawing file		
U_DSN_028	IPE to analyse P&ID drawings to identify: • Control Modules • Process lines • Major Processing Elements	U_US1_005 U_US2_050	
U_DSN_029	• Method of identification will vary by who has drawn the P&ID. Some will include easily identifiable objects with properties, while others will be built up from basic elements. A plug-in defining drawing structure will be required.	U_US1_005	
U_DSN_030	Area model to be organised into an S88 style hierarchy of Area, Process Cell, Unit, Equipment Module, Control Module.	U_US1_011 U_US1_013	
U_DSN_031	IPE to support common resources (equipment not belonging to a particular unit, but available to many)	U_US1_011	
U_DSN_032	IPE to support Shared Resources (Units or Equipment belonging to several parent modules)	U_US1_011	
U_DSN_033	Area Model to be viewable graphically, overlaid on P&IDs	U_US2_049	
U_DSN_034	Area Model to be viewable as a block diagram (including public interface)		
U_DSN_035	Area Model to be viewable in tabular form (similar to explorer)		
U_DSN_036	Area Model to be editable in P&ID view: • Assign modules to parent modules (eg CMs to EMs) • Identify Classes		

U_DSN_037	Area Model to be editable in tabular view
	• Customisable tree structure
	  • Include/exclude P&ID as leaf
	  • Class/Instance view
	  • Assigned/unassigned
	• Assign modules to parent modules (eg CMs to EMs)
	  • Drag and drop to assign
	  • Ctrl D&D to share
	  • Select across instances to identify class
	• Mirror selection in tree view into P&ID view (items selected in tree are highlighted on drawing)		

U_DSN_038	**Module Function Designer**	U_TCH_002	
U_DSN_039	Module Function Designer is where the modules identified in the Area Model have their functionality defined. It covers the range of Function Design (FDS) Procedural Design (PDS) and Detail Design (DDS)		

U_DSN_040	Function designer to specifically include Public Interface design, for defining how modules interact:
	• Module to module interaction type
	  • Master/Slave relationship
	  • Peer relationship
	  • Monitor relationship
	• Command/response definition
	• Fault Monitor status and messaging as part of public interface
		U_US2_013	

U_DSN_041	The IPE Detail Designer to:	U_US2_006	
U_DSN_042	• Contain embedded methodology guidance	U_US1_042 U_US1_070	

<table>
<tr><th>Item</th><th>PCS Design - Requirement</th><th>Source</th><th>Priority (Core, 1, 2)</th></tr>
<tr><td>U_DSN_043</td><td>• Provide graphical SFC editor</td><td>U_US1_058</td><td></td></tr>
<tr><td>U_DSN_044</td><td>• Be prescriptive, enforce standard methodology</td><td>U_US1_060</td><td></td></tr>
<tr><td>U_DSN_045</td><td>• Design at class level, utilising instance mapping table and variances
• Option to design at instance level for single instance modules
• Provide view switching to, view class or any instance
• View Class to Instance mapping as a table
• Include ability to specify instantiation of comments and descriptions eg something like ‘Cream Silo <x|A|B|C>’ would display as ‘Cream Silo x’ in the class, and ‘Cream Silo A’ in instance 1 etc</td><td>U_US1_056
U_US2_090</td><td></td></tr>
<tr><td>U_DSN_046</td><td>• Self configure where possible, eg
  • Infer interlocks from area model
  • Infer ownership from resource assignment.
  • Populate default mode and fault monitors from assigned resources
  • Populate bingo tracks from default settings and assigned resources
  • Validate against design rules (remote device references, CIP/Process interlocks etc)</td><td>U_US1_069
U_US1_071
U_US1_072</td><td></td></tr>
<tr><td>U_DSN_047</td><td>• Support staged evolution of definition:
  • Narrative description of intent,
  • Draft logic with placeholders
  • Complete logic
  • Syntactically correct logic
  • User controls final complete status (logic may be syntactically correct, but incomplete because the user intends to add more to it)
• State of evolution of an objects content to be
  • visible at higher levels (by colour, icon or similar)
  • searchable (eg report all syntactically incorrect transitions)
  • inherited by parent objects (eg a Grafcet cannot be complete if it contains incomplete enables)</td><td>U_US1_075
U_US2_008</td><td></td></tr>
<tr><td>U_DSN_048</td><td>PCS’s typically interface at some point to external out of scope systems.
Function designer to include provision to define interaction to external systems. Some projects allow modification of these external interfaces, some do not. Designer to provide for
• Definition of connection method (eg hard wired, peer to peer comms)
• Definition of connection addressing
• Interface tag list,
• Interface tag function (eg timing diagram, command/response etc)
• Work required in interfaced system (eg None, add comms block, or add IO etc)</td><td>U_US1_092</td><td></td></tr>
<tr><td>U_DSN_049</td><td>Module designer to include prompting and requirement to specify reporting during design phase.
Reporting requirements to be specified at high level and applied to all modules,</td><td>U_US2_066</td><td></td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><td>U_DSN_050</td><td>Visualisation Designer</td><td></td><td></td></tr>
<tr><td>U_DSN_051</td><td>The Visualisation Designer is an environment to assist in the design of HMI and other visualisation components of the PCS.</td><td></td><td></td></tr>
</table>

Item	PCS Design - Requirement	Source	Priority (Core, 1, 2)
U_DSN_052	For HMI design it shall provide: • Scratch pad template (based on client layout requirements) for ‘hand’ sketching overviews and mimics • Base mimic set derived from P&IDs • Ability to reorganise mimics to provide operations centric view (P&IDs are usually laid out from a process centric viewpoint) • Automatic page link identification based on P&ID process lines • Overview designer, high level view defining the set of overviews and their interrelationship • Overview designer, detail view defining the content of the overview pages • Linkage to the Area Model to ensure complete coverage • Menu design • Alarm consolidation and presentation design • Trend page selection • Imbedded object selection (eg Batch objects, report viewers etc)	U_US1_017 U_TCH_002	
U_DSN_053	Visualisation design is independent from Vendor specific requirements. Visualisation Vendor plug-in specifies implementation of the design in a particular platform.		
U_DSN_054	Visualisation design is as much as possible independent from Client specific requirements. Visualisation Client plug-in specifies implementation of the design in a particular platform. There will be some client specific requirements that constrain the design and must be visualised in the designer, such as top and bottom banners, menu structure, etc		
U_DSN_055	**Hardware Architecture Designer**	U_US1_080	
U_DSN_056	Hardware Architecture Designer is where PLC and PC architecture is defined. It includes • PLC hardware selection, IO network design, IO allocation • HMI system selection (servers, clients, redundancy etc) • Batch server system selection (servers, clients, redundancy etc) • MES server system selection (databases, report servers etc) • Support systems selection (engineering work stations, version control servers etc) • Network design		
U_DSN_057	For PLC systems, designer to provide: • Hardware selection and validation (memory capacity, current loadings etc) • Rack location and layout • Redundancy • IO network selection • Peer to Peer comms network • HMI comms network • Specialty networks (eg Profibus, DeviceNet etc) • IO address allocation to devices • System characterisation ie selection of number and size of PLC CPU’s		

Item	PCS Design - Requirement	Source	Priority (Core, 1, 2)
U_DSN_058	For HMI systems, designer to provide: • Hardware selection and validation (memory capacity etc) • Redundancy • Server location • Client location • Network selection • HMI comms network • Specialty networks (eg Profibus, DeviceNet etc) • System characterisation ie selection of number and size of servers		
U_DSN_059	**Plug-Ins**		
U_DSN_060	Plug-ins are required to support the mapping of client and vendor specific requirements to the CTEK standard models and methods		
U_DSN_061	Project base templates may be modified within constraints laid down by the client standards for the project at hand	U_US2_077	
U_DSN_062	*Input documents*	U_US2_048 U_US2_052	
U_DSN_063	*Drawings*		
U_DSN_064	Support for dwg and dxf file format		
U_DSN_065	Specify how client drawing format constructs module objects and how/if they contain object properties (eg tag name)		
U_DSN_066	*Schedules*		
U_DSN_067	Support for database, xls, doc, xml, csv, txt formats		
U_DSN_068	Specify how client document lays out module properties		
U_DSN_069	*Specifications*		
U_DSN_070	Support for database, doc, xml, csv, txt formats		
U_DSN_071	Specify how client document format lays out requirements		
U_DSN_072	*PCS documentation*		
U_DSN_073	Support for MSOffice formats (xls, doc, vsd, ppt)		
U_DSN_074	Specify mapping of CTEK PCS model to client document layout		
U_DSN_075	*Area Model*		
U_DSN_076	*FDS*		
U_DSN_077	*PDS*		
U_DSN_078	*DDS*		
U_DSN_079	*HRS*		
U_DSN_080	*Change Control Records*		
U_DSN_081	*PLC Platform*		
U_DSN_082	Plug-in provides platform specific interfaces to a particular vendors PLC. Eg for ControlLogix plug-in provides scrap file, l5k, l5x, tag list csv file formats		

Item	PCS Design - Requirement	Source	Priority (Core, 1, 2)
U_DSN_083	*PLC Code*		
U_DSN_084	Plug-in provides mapping of IPE PCS model components to PLC code elements, including file structure, tags, comments, logic elements Eg for ControlLogix plug-in would include Tags: scope, name, type, description, alias, value File structure: Tasks, Programs, Routines, including config and descriptions Logic: routine type (eg ladder), logic, rung comment IO tree: card type, location, name, comment, config		
U_DSN_085	Plug-in to include a template project file containing only the standard components present in all projects. This form the starting point for all PLC project files.	U_US1_076	
U_DSN_086	*HMI*		
U_DSN_087	Plug-in to include a template project file containing only the standard components present in all projects. This form the starting point for all HMI project files.	U_US2_076	
U_DSN_088	*Batch*		
U_DSN_089	Plug-in to include a template project file containing only the standard components present in all projects. This form the starting point for all Batch project files.	U_US2_076	
U_DSN_090	*MES*		
U_DSN_091	*Reporting tags*		
U_DSN_092	*Reports*		
U_DSN_093	*Track and Trace*		
U_DSN_094	**Change Implementation Assistant**		
U_DSN_095	Once the project has reached a stage where implementing changes requires steps external to the IPE, making a change will generate a task list required to implement the change. The extent of the task list will depend to some extent on the implementation state of the project.	U_US1_038	
U_DSN_096	The change process shall generate a set of files necessary to implement the change, eg scrap files for adding/modifying ControlLogix code, tag import files, tag description import files, tag values for download, sql update queries, hmi import files, instructions for manual changes where automated application is not possible. The form and content of files generated to be appropriate to the project stage, (eg offline vs online) and client and vendor plug-ins in use	U_US1_090 U_US2_043	
U_DSN_097	If the project is at the stage where site change control procedures apply, a conforming change control document shall be produced.		
U_DSN_098	The task list shall require sign off of each step by the engineer implementing the change		

Item	PCS Design - Requirement	Source	Priority (Core, 1, 2)
U_DSN_099	This assistant shall include support for system input changes, such as P&ID revisions, addition or deletion of IO Adding a device would trigger: • Module allocation requirement • Module definition changes: additions to sequence logic, tracks, enables • Visualisation changes for new module (mimic, overviews) • Alarm definition changes • Device ID allocation • Impact on other systems (eg MES, Batch etc) Adding a device would generate a task list, including • Module allocation (process and or CIP) • HMI tag import file • Mimic to update (based on P&ID location) • Prompt to consider other HMI changes (eg overview, or alarm summary) • HMI graphic import for symbols (if possible) • PLC tag import file (including device tag, io tags, alarms, module tracks etc) • Device logic import file • Tag values download • EM and Phase logic import file (transition logic, ownership, monitors, outputs) • Co-ordination logic import file • Alarms logic import file • Message database update sql script • Updates for other systems (eg MES, Batch etc)	U_US2_044 U_US2_047	

5.7. PCS Construction

Requirements related to the implementation of a PCS

Item	PCS Construction - Requirement	Source	Priority (Core, 1, 2)
U_CST_001	**Code Generation**		
U_CST_002	IPE to include generation of PLC code from PCS model, utilising vendor plug-in (specifies target PLC platform) and client code plug-in (specifies code layout)	U_US2_078	
U_CST_003	Ideally generator to produce 100% of code. Failing that any manually created code to be identifiable, and fed back to IPE to facilitate task list generation for changes and to protect and preserve manually created code.	U_US1_078 U_US2_067	
U_CST_004	**On Site Work**		
U_CST_005	IPE to be used throughout project life cycle, including during installation and commissioning	U_US1_091 U_US2_035	
U_CST_006	Module status tracking to extend through installation and commissioning stages, tracking states of: • Installing (including IO checkout in progress) • Installed (ready to dry test) • Dry Testing • Dry Tested • Wet Testing • Wet Tested • Product Testing • Ready for Takeover • Site Control (post takeover) • Ready for Acceptance (all takeover hit list items complete) • Accepted	U_US1_091 U_US2_035	
U_CST_010	Above states list is liquid process focused. IPE to support states appropriate to a variety of industries	U_US3	
U_CST_007	Once an item has reached Takeover status, any change are to be authorised and reported using client specified Change Control practices	U_US1_091 U_US2_035	

5.8. PCS Testing

Requirements related to the testing of a PCS, at all stages of development

Item	PCS Testing - Requirement	Source	Priority (Core, 1, 2)
U_TST_001	**Early Stage Testing**		
U_TST_002	Module designer to include Public Interface simulation function to test module to module interaction. This simulation to occur in the IPE, without PLC code. Aims of the test is to: • Exercise command/response • Detect command deadlocks • Exercise module to module messaging	U_US1_022 U_US2_013 U_TCH_008	
U_TST_003	Module designer to include Grafcet simulation function to test module operation. Test to be carried out at Class level or at Instance level This simulation to occur in the IPE, without PLC code. Aims of the test is to: • Exercise module sequencing • Exercise module response to commands • Exercise module exception handling • Exercise module device control	U_US1_020 U_US1_022 U_US1_057 U_TCH_008	
U_TST_004	Visualisation Designer to include simulation function to animate module state on mimic and overview mock-ups. This simulation to occur in the IPE, without PLC or HMI code. Aims to the test is • Early demonstration of procedural and equipment control decisions • Exercise operator interaction with the system	U_US1_018 U_US1_019 U_US2_059 U_TCH_008	
U_TST_005	**Module Testing**		
U_TST_006	CTEK standard methodology to include standardised module test methodology. The IPE shall: • Generate module test plan • Monitor test execution and record results • Track test plan progress, and release module on test pass • Generate test reports • Automatically execute module tests where possible	U_US1_022 U_US1_087 U_US1_088 U_US1_089 U_US2_034 U_US1_056 U_US1_057 U_TCH_008	
U_TST_007	Module testing to be possible with or without HMI (perhaps use Visualisation Designer to animate module behaviour)	U_US2_058 U_TCH_008	
U_TST_008	**Integration Testing**		
U_TST_009	CTEK standard methodology to include standardised integration test methodology. The IPE shall: • Generate integration test plan • Monitor test execution and record results • Track test plan progress, and release modules on test pass • Generate test reports	U_US1_022 U_TCH_008	

Item	PCS Testing - Requirement	Source	Priority (Core, 1, 2)
U_TST_010	Integration testing is based on running PLC code and HMI in simulation mode. In addition to built in simulation logic in the modules, the IPE shall: • Have ability to capture and restore a process state to recreate an initial condition (perhaps by upload/download of all relevant tag values to be able to set a group of modules to a known state • Have to ability to manipulate process inputs in concert with tests, eg set proximity switch state on certain steps, manipulate an analogue signal in response to a process condition (represents basic process simulation)	U_US1_021 U_US1_022 U_US2_092 U_TCH_008	
U_TST_011	**Factory Acceptance Testing**		
U_TST_012	CTEK standard methodology to include standardised FAT test methodology. The IPE shall: • Generate FAT test plan • Monitor test execution and record results • Track test plan progress, and release modules on test pass • Generate test reports	U_US2_034 U_TCH_008	
U_TST_013	**Site Acceptance Testing**		
U_TST_014	CTEK standard methodology to include standardised SAT test methodology. The IPE shall: • Generate SAT test plan • Monitor test execution and record results • Track test plan progress, and release modules on test pass • Generate test reports	U_TCH_008	

Appendix I. Workshop #1 – User Requirements

A series of workshops and meetings have been held to gather requirements from a user's perspective. The following are the requirements as record during those events. These requirements will be analysed, qualified and distilled into the IPE User Requirement.

WS01 = Workshop #1 10/11/2010 – TJB, PSL, Ash, Anton, JDH, LJM, CJN, GAM

Item	Requirement	Source	Priority (Core, 1, 2)	XRef
U_US1_001	A standard Project Lifecycle Methodology is required.	WS01		U_GP_002
U_US1_002	A single master repository is required to store all the elements/data for a PCS's definition. [Data only in one place]	WS01		U_GP_006
U_US1_003	Capture or include the costing and tendering data into the system (repository). Including but not limited to: • P&IDs • URS and scope documents • IO • Panel layouts • Architecture • Process Flow drawings • Area Model information/drawings. • Client standards • Photos and Video	WS01		U_TND_003
U_US1_004	The IPE is to facilitate handover from the proposal team to the execution team.	WS01		U_PM_017
U_US1_005	Electronically capture modelling information from P&IDs into the ENVIRONMENT.	WS01		U_DSN_028 U_DSN_029
U_US1_006	Capture tender/design notes, decisions and considerations into the ENVIRONMENT.	WS01		U_TND_003
U_US1_007	Capture costing information into the ENVIRONMENT.	WS01		U_TND_003
U_US1_008	Data to reside only in one place in the ENVIRONMENT. Data only to be entered once into the ENVIRONMENT.	WS01		U_GP_006
U_US1_009	Data to only be in one place.			U_GP_005
U_US1_010	All data/elements for a Project are to contain revision information and history.	WS01		U_GP_007
U_US1_011	The design environment is to support a Hierarchal genealogy of the elements of the design. Forward and backward traceability is to be provided and inherent in the system.	WS01		U_DSN_030 U_DSN_031 U_DSN_032
U_US1_012	Familiarisation of the team with the site and process is to be part of the project methodology.	WS01		U_PM_001
U_US1_013	All data is to have context.	WS01		U_DSN_030
U_US1_014	A standard way/method of how we go about automating plant/items is to be defined and documented.	WS01		U_GP_003
U_US1_015	Must be able to draw on/from previous works.	WS01		
U_US1_016	The design environment needs to be independent of the client's standards.	WS01		U_DSN_001
U_US1_017	We require a HMI designer that is non platform specific and independent of client specific standards/requirements.	WS01		U_DSN_052
U_US1_018	HMI views able to support early stage testing/simulation. Specific implementation detail not required to support this.	WS01		U_TST_004

Item	Requirement	Source	Priority (Core, 1, 2)	XRef
U_US1_019	System to support early design stag(s) testing and/or design verification.	WS01		U_TST_004
U_US1_020	Would be nice to test the sequencing at detail design stage.	WS01		U_TST_003
U_US1_021	Easy generation of basic device feedback and basic process plant simulation "code".	WS01		U_TST_010
U_US1_022	Require simulation capability at Functional Specification (FS), Detailed Design and Implementation stages.	WS01		U_TST_002 U_TST_003 U_TST_006 U_TST_009 U_TST_010
U_US1_023	Require Hierarchical design environment and views. If something in one level is changed then it automatically flows through to all other levels.	WS01		U_GP_005
U_US1_024	Area Modelling – describes the process/method of breaking up the process plant to be automated.	WS01		U_DSN_025
U_US1_025	A standard modelling methodology shall be defined and documented. This shall include standard terminology.	WS01		U_GP_003
U_US1_026	Training and training materials are required to support the use of the ENVIRONMENT.	WS01		U_FTR_009 U_FTR_011
U_US1_027	In built "help", tips, and knowledgebase. A Wiki type environment is required to allow users freedom to contribute knowledge and experiences.	WS01		U_FTR_019
U_US1_028	Libraries of information shall be a feature at all level.	WS01		U_DSN_023
U_US1_029	Templates shall be a feature of the ENVIRONMENT.	WS01		U_FTR_038 U_FTR_039
U_US1_031	Wizards to help you through the modelling process.	WS01		
U_US1_032	The system is to be independent of client documentation requirements (Output).	WS01		U_FTR_070
U_US1_033	The environment is to be "Integrated".	WS01		U_UI_002
U_US1_034	Need to be able to bring work back into the system.	WS01		U_FTR_055
U_US1_035	Work must be able to be resynchronised to the office master.	WS01		U_FTR_056
U_US1_036	A design environment is required.	WS01		U_DSN_007
U_US1_037	All aspects of the system are to be easy to use.	WS01		U_GP_009
U_US1_038	Task lists for changes shall be generated automatically dependent on a projects stage.	WS01		U_DSN_095
U_US1_039	Ability to bring in an existing PCS's set of implementation files and then support those systems is required. The Open Country Cheese situation is an opportunity for this requirement. This could be a service that we offer when supporting sites.	WS01		U_FTR_045
U_US1_040	It shall be more productive to do our work in the ENVIRONMENT than current methods.	WS01		U_GP_001
U_US1_041	One package. As a minimum the look and feel of a single integrated Package.	WS01		U_UI_002
U_US1_042	Embedded design method/process guidance.	WS01		U_DSN_042
U_US1_043	Con-current multi-user views of all data.	WS01		U_FTR_021
U_US1_044	Graphical /drill down	WS01		U_UI_001
U_US1_045	Filters	WS01		U_UI_005
U_US1_046	Consistent menus, quick keys etc	WS01		U_UI_004
U_US1_047	Consistent well designed layout.	WS01		U_UI_004
U_US1_048	Graphical User Interface required.	WS01		U_UI_001
U_US1_049	Security is required.	WS01		U_GP_010 U_FTR_050

Item	Requirement	Source	Priority (Core, 1, 2)	XRef
U_US1_050	The system is to be able to be used from (connected to the master or not) • Office • Remote office • Home • Client site	WS01		U_GP_011 U_GP_012 U_GP_014
U_US1_051	Help and User manuals are required.	WS01		U_FTR_014
U_US1_052	Multi-user and collaborative environment is required.	WS01		U_GP_010 U_GP_011
U_US1_053	Items or aspects shall be locked to a single user as appropriate.	WS01		U_FTR_023
U_US1_054	Cross notification when more than one user accessing an item.	WS01		U_GP_013 U_FTR_024 U_FTR_027
U_US1_055	Cross referencing and searching capability required.	WS01		U_UI_006 U_UI_007 U_UI_008
U_US1_056	Class level design approach required. Then to instances.	WS01		U_DSN_045 U_TST_006
U_US1_057	Class level testing is required.	WS01		U_TST_003 U_TST_006
U_US1_058	Graphical SFC editor for the Detailed Design Environment (DDE)	WS01		U_DSN_043
U_US1_059	DDE to be contextually aware and smart.	WS01		
U_US1_060	DDE environment to be prescriptive.	WS01		U_DSN_044
U_US1_061	Components in development in the DDE to have a status.	WS01		U_PM_006
U_US1_062	Every item in the system is to have a “Unique Identifier”	WS01		U_DSN_018
U_US1_063	Allow use of ladder design logic in the DDE. And view.	WS01		U_UI_009
U_US1_064	Give a visual high light of items/aspects that are not complete.	WS01		U_UI_010
U_US1_065	Be able to provide the client with the minimum documentation that they require.	WS01		U_FTR_072
U_US1_066	Administration and management of the environment is to be supported.	WS01		U_PM_001
U_US1_067	An environment to build, load, configure the clients requirements for there specific platform and/or standard is required.	WS01		U_FTR_004
U_US1_068	Access to the client requirements functionality shall be controlled.	WS01		U_FTR_031
U_US1_069	Detailed design verification functionality is required; aspects such as peer to peer, remote device and phase to phase interlock.	WS01		U_DSN_046
U_US1_070	DDE to guide you through the process of doing the detailed design.	WS01		U_DSN_042
U_US1_071	Automated ownership definition and verification.	WS01		U_DSN_046
U_US1_072	Standard components (elements, phase steps for example) of the detailed design to come by default.	WS01		U_DSN_046
U_US1_073	Templates, slices and chunks of components to be supported. Ability to gear at many levels of complexity and/or scope.	WS01		U_FTR_039
U_US1_074	Be able to bring in clients input documentation i.e. FD’s many things and many different stages.	WS01		
U_US1_075	Utilise modular implementation/design/development methods of the ENVIRONMENT to provide flexibility and staged implementation of functionality.	WS01		U_DSN_047
U_US1_076	Create platform specific base specific base configuration for HMI, PLC and Batch.	WS01		U_DSN_085

Item	Requirement	Source	Priority (Core, 1, 2)	XRef
U_US1_077	Be able to update, modify and change platform base configurations.	WS01		
U_US1_078	Automatically be able to generate all items that have been defined in the system to the appropriate platform specific files.	WS01		U_CST_003
U_US1_079	System to be able to handle multiple logic implementation methods.	WS01		U_DSN_002
U_US1_080	System to provide detailed design of : • Architecture • Rack layout etc • Networks, etc All aspects of a PCS hardware design and implementation.	WS01		U_DSN_055
U_US1_081	All outputs/views to include/display version and status.	WS01		U_UI_011
U_US1_082	Provide reviewer mode/distribution for external review. [Post meeting discussion suggests that the ability to provide access into the office system over the internet may be the most cost effective way to provide this functionality. We may make this the specific requirement for this request.	WS01		U_FTR_030
U_US1_083	Customer and Peer review comments and feedback to be recorded/versioned into the system for future reference/review. It is to persist.	WS01		U_FTR_035
U_US1_084	Fault and commissioning fault logs in the system.	WS01		U_PM_015
U_US1_085	Project progress monitoring to be inherent to the system.	WS01		U_PM_006 U_PM_007 U_PM_010
U_US1_086	Assignment of tasks to resource. Task list. Status of tasks against the list.	WS01		U_TST_006
U_US1_087	Automatic generation of test cases for all phases of the project (V Model)	WS01		U_TST_006
U_US1_088	Record testing.	WS01		U_TST_006
U_US1_089	Verification of testing progress/outcome.	WS01		
U_US1_090	Automatic generation of work lists depending on stage and task to be done.	WS01		U_PM_010 U_DSN_096
U_US1_091	Site changes could be under site change control or not. Need to be able to cater for these types of steps in the work lists.	WS01		U_CST_005 U_CST_006 U_CST_007
U_US1_092	Need to be able to deal with non direct project scope interfaces, i.e. remote PLCs	WS01		U_DSN_048
U_US1_093	Scalable. We require the system to be able to be used efficiently on projects that range from being small to very large.	WS01		U_GP_011 U_GP_015
U_US1_094	Project start up effort to be very efficient, such that the barrier to use it on a small project is minimal.	WS01		U_GP_015
U_US1_095	Develop method for analysing the applicability of the system to each specific project. Utilisation of this should be part of the tender process.	WS01		U_TND_002
U_US1_096	Project takeover and acceptance stages to be fully supported.	WS01		U_PM_037 U_PM_038
U_US1_097	Need to be able to download and upload data to/from deployment platform.	WS01		U_FTR_105
U_US1_098	Consider client IP protection.	WS01		U_FTR_053
U_US1_099	Project Management to be integral to the system.	WS01		U_PM_001
U_US1_100	Quality Assurance to be integral to the system.	WS01		U_PM_034

Item	Requirement	Source	Priority (Core, 1, 2)	XRef
U_US1_101	*Unfortunately I did not capture this item – can anyone remember???*	WS01		
U_US1_102	Live view of status and data in design view when connected to the implemented system. [Post meeting addition by GAM]	WS01		U_UI_013
U_US1_103	The ENVIRONMENT shall be able to handle DDS in Grafcet excel format from clients or previous project work. [Post meeting addition by GAM]	WS01		U_FTR_048
Note:	[The group estimated that an environment that satisfied the above requirements may reduce the effort required to execute the works by around 50% across the design, coding and office testing stages (around 50% of the overall project work effort). Overall 25% net reduction in project effort.]			

Appendix II. Workshop #2 - User Requirements

WS02 = Workshop #2 24/11/2010 – JEH, TJB, CJN, GAM, Emerson, Jason, Tony G

Item	Requirement	Source	Priority (Core, 1, 2)	XRef
U_US2_001	Structured approach to modelling the plant. Standard documented methodology.	WS02		U_PM_032
U_US2_002	Standard method (documented) to execute the work.	WS02		U_GP_003
U_US2_003	Standard project management /team meetings on projects.	WS02		U_GP_002
U_US2_004	System to be independent of the deploy platform.	WS02		U_PM_005
U_US2_005	Modular. Able t add components/platforms/clients in a modular way.	WS02		U_FTR_001
U_US2_006	Necessary to have a Detailed Design step.	WS02		U_FTR_001
U_US2_007	V- model approach – Top down design and Bottom up testing.	WS02		U_DSN_041
U_US2_008	URS/FS – flowchart /SFC/Words	WS02		U_PM_035
U_US2_009	Having client signoff of what going to deliver important. Needs to be documented.	WS02		U_DSN_047
U_US2_010	System to support V-model approach	WS02		U_PM_018
U_US2_011	Design notes captured in the system.	WS02		U_PM_035
U_US2_012	Different levels of reporting for internal and external built into the system.	WS02		U_DSN_021
U_US2_013	Design environment separate from how you want to present the information to the client. [Jason mentioned a product he had come across "Yourdon(sp?)]	WS02		U_PM_019
U_US2_014	Collaborative design/build environment.	WS02		U_FTR_001 U_DSN_040 U_TST_002
U_US2_015	CTEK way of building/designing automation requirements required.	WS02		U_FTR_020
U_US2_016	Library of components/templates at all levels of size and complexity.	WS02		U_GP_002
U_US2_017	Menu/Drop down to get to components.	WS02		U_FTR_038
U_US2_018	Environment to have rules/guide you through the process.	WS02		U_UI_004
U_US2_019	Gates for peer reviews at all levels of the v-model.	WS02		U_DSN_024
U_US2_020	Scalability of the size of the project that can be executed in the environment	WS02		U_PM_035
U_US2_021	Support training of all users. New and experienced in automation.	WS02		U_GP_015
U_US2_022	Want to capture decisions and assumptions from Initiate to Close-out. To persist through the environment and the steps of the project.	WS02		U_FTR_008 U_FTR_012
U_US2_023	Project management to be integral to the environment.	WS02		U_TND_004
U_US2_024	Environment to support project management requirements.	WS02		U_PM_001
U_US2_025	Want changes to propagate automatically to all levels of design / implementation.	WS02		U_PM_001
U_US2_026	Enter data once in one place.	WS02		U_GP_005
U_US2_027	Single repository for the definition information for the project and Process Control System design.	WS02		U_GP_005

Item	Requirement	Source	Priority (Core, 1, 2)	XRef
U_US2_028	Ability to gear off previous work - legacy - new environment	WS02		U_GP_006
U_US2_029	Design/Decision notes to persist against the design model.	WS02		U_TND_003 U_FTR_033 U_DSN_021
U_US2_030	Client IP needs to be protected.	WS02		U_TND_004
U_US2_031	Want environment to be independent of client and deployment platform	WS02		U_FTR_053
U_US2_032	Client documentation and requirements are an input to the system. Not the design environment.	WS02		U_FTR_001
U_US2_033	Support quality.	WS02		U_FTR_001
U_US2_034	Promote good testing and FAT.	WS02		U_GP_003
U_US2_035	System to be used at site also.	WS02		U_TST_006 U_TST_012
U_US2_036	Record parameter and tuning in the system. [Consider potential IP issues with our customers]	WS02		U_CST_005 U_CST_006 U_CST_007
U_US2_037	Change and revision control on all aspects and stages including site work.	WS02		U_FTR_053
U_US2_038	Verbal control of design process. {No limits!}	WS02		U_GP_007 U_FTR_069
U_US2_039	Drag and drop functionality.	WS02		U_FTR_029
U_US2_040	Short cut keys. Common navigation etc	WS02		U_UI_004
U_US2_041	Graphical User Interface {GUI}	WS02		U_UI_004
U_US2_042	Change data in one place.	WS02		U_UI_001
U_US2_043	Produce change lists/files dependent on stage of project. Automatically produced.	WS02		U_GP_005
U_US2_044	Additions of IO or functionality gets context that can produce task lists to include.	WS02		U_DSN_096
U_US2_045	Generate incomplete work lists and status.	WS02		U_DSN_099
U_US2_046	Generate and maintain work item status.	WS02		U_PM_020
U_US2_047	Needs to handle changing Inputs. i.e. PID, IO, sequence etc	WS02		U_PM_006
U_US2_048	Be able to import PID's and all inputs.	WS02		U_DSN_099
U_US2_049	Use imported PID's to overlay area model design.	WS02		U_PM_002 U_DSN_062
U_US2_050	Automatically identify objects on the PID's.	WS02		U_DSN_033
U_US2_051	PID's to be updated and verified at the start of a project. CTEK methodology for all projects. Master input.	WS02		U_DSN_028
U_US2_052	Need to be able to bring in varying initial project inputs - PID's - IO - URS - Written descriptions - [careful about building dependency's]	WS02		U_PM_002
U_US2_053	Drill down and up through the design levels.	WS02		U_DSN_062
U_US2_054	Be able to cross reference in the design environment.	WS02		U_UI_004
U_US2_055	Automatically generate requirements matrix and traceability.	WS02		U_UI_006
U_US2_056	Auto generate test sheets and auto/manual record outcomes in the system. From design and definition information.	WS02		U_PM_012 U_PM_029
U_US2_057	Automated testing. - level to be determined - basic functional?	WS02		

Item	Requirement	Source	Priority (Core, 1, 2)	XRef
U_US2_058	Be able to test prior to HMI implemented into deploy platform.	WS02		
U_US2_059	Be able to demonstrate at FS level onto PID to convey/capture intent with client.	WS02		U_TST_007
U_US2_060	Be able to record information from demonstration at FS level for future reference. And the points/issues etc raised/identified.	WS02		U_TST_004
U_US2_061	Give the client the minimum that they require.	WS02		U_FTR_032
U_US2_062	Design environment to be independent of implementation.	WS02		U_FTR_001
U_US2_063	Specific implementation requirements/standards for a client/platform are a implementation requirement only. Not to dictate the design environment.	WS02		U_FTR_001
U_US2_064	Need to be able to cater for client specifics we deploy. Need to be able to build and maintain client and platform specifics.	WS02		U_FTR_001
U_US2_065	For PLC, HMI, Batch and MES want to be able to deploy to multiple platforms and client requirements.	WS02		U_FTR_001
U_US2_066	Reporting and other MES requirements integral through the process. Not tack on's!	WS02		U_FTR_001
U_US2_067	Automatically generate all deploy platform specific configuration files.	WS02		U_DSN_049
U_US2_068	Efficient and User friendly environment.	WS02		U_CST_003
U_US2_069	Training supported and inherently in the system.	WS02		U_GP_009
U_US2_070	Training before you use the system.	WS02		U_FTR_009
U_US2_071	Standard user shortcuts.	WS02		U_FTR_009 U_FTR_010
U_US2_072	Different views of the environment and steps to be supported dependent on user group / view requirement.	WS02		U_UI_004
U_US2_073	User group customisation	WS02		U_UI_014
U_US2_074	A level of user customisation in the GUI.	WS02		U_UI_014
U_US2_075	Security – user login etc	WS02		
U_US2_076	Management of users/groups for the specific project.	WS02		U_GP_010 U_FTR_050
U_US2_077	Administration of the environment.	WS02		U_PM_025 U_DSN_085 U_DSN_087 U_DSN_089
U_US2_078	Need to protect CTEK IP and environment IP. Dongle/License??	WS02		U_FTR_050 U_FTR_051 U_DSN_061
U_US2_079	Integration with email/Outlook and other business systems; - SVN - Network - Wiki - Mindmaps - SB1 - Workbench - etc	WS02		U_GP_010 U_CST_002
U_US2_080	Identify all team members and allocate tasks to the team members.	WS02		U_PM_005 U_FTR_054 U_FTR_069
U_US2_081	Risk management and mitigation process to be integral.	WS02		U_PM_010 U_PM_025
U_US2_082	Record and identify all internal and external project people. Their contacts and their roles in the project.	WS02		U_PM_013
U_US2_083	Integrate with overall resource availability and assignment in CTEK.	WS02		U_PM_004

Item	Requirement	Source	Priority (Core, 1, 2)	XRef
U_US2_084	Project lifecycle and control gates with appropriate authority level.	WS02		
U_US2_085	Status not only by component but down to the elements within the components. [Terminologies need to be defined early for this development so that we all know what we are talking about]	WS02		U_PM_027
U_US2_086	Reporting of status against components and assigned resource.	WS02		U_PM_008
U_US2_087	Revision history at all levels for all actions.	WS02		U_PM_020
U_US2_088	Environment to assist in identifying bottlenecks in the project.	WS02		U_GP_007
U_US2_089	System to be able to build step based systems.	WS02		U_PM_011 U_PM_014
U_US2_090	Class based design/implement methodology.	WS02		U_DSN_003
U_US2_091	Livening up the design environment for testing and commissioning.	WS02		U_DSN_045
U_US2_092	Ability to create a test scenario and be able to automatically get to the start state for the test . The start state could be at the first step or somewhere through the test process.	WS02		U_UI_013
U_US2_093	Parameter/Tuning and configuration upload/download.	WS02		U_TST_010
U_US2_094	CTEK have a standard for HMI, PLC and Batch for clients that give you a choice.	WS02		
U_US2_095	The environment needs the ability to define dependencies in order to define where to start detailed design, coding and testing. This dependency should also take into account classes to lift priority. [Post meeting item TJB]	WS02		U_FTR_007
				U_PM_011 U_PM_014 U_PM_030
U_US2_096	While conducting meeting, the IPE should be available for direct entry of minutes, thoughts and comments A component could exist as a meeting record (recording when where why and who), with minutes including contextual links to related parts of the model [Post meeting item CJN]	WS02		U_FTR_028
U_US2_097	Use of a data board (image projected onto an intelligent board, use your finger as a electronic pen) for group modelling exercises, and (remember, no boundaries) voice activated commands [Post meeting item CJN]	WS02		U_FTR_029
Note:	[The group estimated that an environment that satisfied the above requirements may reduce the effort required to execute the works by around 20%]			

Appendix III. Technical Group - User Requirements

Item	Requirement	Source	Priority (Core, 1, 2)	XRef
U_TCH_001	The user interface is to be Graphical in form. GUI.	Technical		
U_TCH_002	Designer Tools. • Area Model Designer Tool • Functional Designer Tool • Detailed Designer Tool • Visualisation Designer Tool • Plug-In Designer Tool • Template Designer Tool	Technical		U_DSN_007 U_DSN_038 U_DSN_025 U_DSN_052
U_TCH_003	Editor Tools. • Tag Editor Tool • Sequential Function Chart (SFC) Editor • Procedural Function Chart (PFC) Editor • Bingo Editor Tool • Enable Editor Tool • Fault/Monitor Mode Editor Tool • Resource Editor Tool	Technical		
U_TCH_004	Definition Tools. • Module Definition Tool • Plug-in Definition Tool • PLC Platform Plug-in Definition Tool • HMI Platform Plug-in Definition Tool	Technical		
U_TCH_005	Viewing Tools	Technical		
U_TCH_006	Reporting Tools. • Standard Model Definition Reports • Functional Design Specification Report • Detailed Design Specification Report • Project management Reports • Client Specific Reports • Client Specific Area Model Report • Client Specific Functional Design Specification Report • Client Specific Detailed Design Specification Report	Technical		U_PM_020 U_PM_023
U_TCH_007	Manager Tools. • Template Manager Tool • Previous Work Manager Tool	Technical		
U_TCH_008	Testing Tools.	Technical		U_TST_002 U_TST_003 U_TST_004 U_TST_006 U_TST_007 U_TST_009 U_TST_010 U_TST_012 U_TST_014
U_TCH_009	Verification Tools	Technical		
U_TCH_010	Simulation Tools.	Technical		
U_TCH_011	Code Generation Tools. • Logic code generation tool • Visualisation code generation tool	Technical		
U_TCH_012	Diagnostic Tools.	Technical		U_GP_019

Item	Requirement	Source	Priority (Core, 1, 2)	XRef
U_TCH_013	Training and Training Manuals	Technical		U_FTR_008
U_TCH_014	Previous Work.	Technical		U_FTR_045
U_TCH_015	Administration	Technical		U_FTR_049 U_FTR_050 U_FTR_051
U_TCH_016	Efficiency for CTEK users.	Technical		U_GP_009
U_TCH_017	Independence - Client, Industry and Vendor	Technical		U_FTR_001
U_TCH_018	Architecture to provide scalable and distributable infrastructure	Technical		U_GP_011 U_GP_012
U_TCH_019	Collaboration	Technical		U_GP_010 U_GP_013
U_TCH_020	Platform specific deployment files plug in	Technical		U_FTR_002
U_TCH_021	Platform specific implementation requirement plug in	Technical		U_FTR_003
U_TCH_022	Customer specific requirements plug in.	Technical		U_FTR_004
U_TCH_023	Customer specific input and output plug in.	Technical		U_FTR_005
U_TCH_024	CTEK Preferred inputs and outputs	Technical		U_GP_017
U_TCH_025	Help and Support	Technical		U_FTR_014 U_FTR_015 U_FTR_016 U_FTR_017 U_FTR_018
U_TCH_026	Business System Interfaces/Integration • SB1 • Best Practice including CSIA and Quality systems • MindMap • Microsoft Outlook • SVN • Wiki Knowledge base • Timesheet • Workbench	Technical		U_GP_008 U_PM_005 U_FTR_052 U_FTR_054 U_FTR_069
U_TCH_027	Repositories • IPE Definition Repository • Current Project Definition Repository • Previous Work Metric's and KPI Repository • Previous Work Repository	Technical		U_FTR_067
U_TCH_028	Change Control	Technical		U_GP_008 U_FTR_069
U_TCH_029	Excel Grafcet environment	Technical		U_GP_018
U_TCH_030	Standard Language and Terminology	Technical		U_GP_016
U_TCH_031	Standards	Technical		U_GP_003
U_TCH_032	Projecting Methodology	Technical		U_GP_002 U_GP_003
U_TCH_033	Object Model(s)	Technical		U_FTR_006
U_TCH_034	Project Activity Model	Technical		U_GP_003
U_TCH_035		Technical		
U_TCH_036		Technical		